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Studierichting: Informatica
Oefeningenreeks 4A nr. 19.1

$$I = \int e^x \sin(x)dx$$

$$\left\{ \begin{array}{l} u = \sin(x) \\ dv = e^x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \cos(x)dx \\ v = e^x \end{array} \right.$$

$$I = e^x \sin(x) - \int e^x \cos(x)dx$$

$$J = \int e^x \cos(x)dx$$

$$\left\{ \begin{array}{l} u = \cos(x) \\ dv = e^x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = -\sin(x)dx \\ v = e^x \end{array} \right.$$

$$J = e^x \cos(x) - \int e^x (-\sin(x))dx$$

$$J = e^x \cos(x) + \int e^x \sin(x)dx$$

$$J = e^x \cos(x) + I$$

$$I = e^x \sin(x) - J$$

$$I = e^x \sin(x) - (e^x \cos(x) + I)$$

$$I = e^x \sin(x) - e^x \cos(x) - I$$

$$2I = e^x \sin(x) - e^x \cos(x) + C$$

$$I = \frac{e^x}{2} (\sin(x) - \cos(x)) + C$$

References